create and modify them. You don't have to draw them but produce them using code. With this you'll get an idea of how the computer draws shapes and manipulates them behind the scenes.

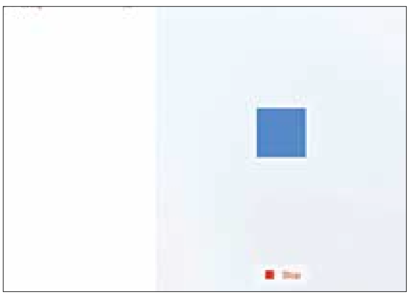
Once you open the template, a pre-generated code will be shown, which is an example of 'how to draw a circle'. Here's the explanation of the code you see.

```
let circle = Circle()
```

Here you have created a variable circle and assigned a constructor named Circle(). If you are a programming novice, you won't know what a constructor is. For simplicity, consider Circle() as a method that will draw a circle on the screen.

```
circle.draggable = true
```

Since your variable handles the circle you can use its name to manipulate your circle every time. For instance, in circle.draggable the first part "circle" is our variable name and "." period is used to access its properties. In the above example, draggable



Modify code to draw a rectangle.

is a property that allows you to drag the circle across the screen. If you set the property to "true", you'll be able to drag the circle.

You can replace the constructor Circle() with Rectangle() and see how your code runs. With this you can draw different shapes and manipulate them.

Draw your name in a fancy way

Write the following code:

```
let text = Text(string:
"Digit", fontSize: 48.0, font-
Name: "Futura", color: .blue)
text.center.y -= 2
```

Here we create a variable name text and assign the Text() constructor to it. To make it fancy, however, we add some attributes to this constructor, which is different from a method. So here, you see we haven't directly entered the name within the parenthesis, instead entering different attributes, which are responsible for changing the appearance of the text.

Your final result should look like this.

The first attribute string: allows you to enter the text to be displayed. In the above example, we have entered Digit within the double quotes. You may change it to anything you want. The second attribute fontSize: lets you specify the text size. The third attribute fontName: allows you to specify the font. Lastly, color: attribute sets a colour. You may change one or more attributes to know how this thing works.

Blank Canvas

The Blank Canvas is not restricted to Shapes or Answers and gives you a free hand. You could start with raw coding here; try different methods out and truly explore programming.

A similar popup will appear showing your name.

Start by displaying your name with a different method than show(). Make use of the standard programming method - print - to display text on the screen. Write:

```
print("Your Name")
```

Hit the "Run my code button". Your code will run in a separate popup window showing whatever you have written. Below the popup you'll see a button "Add viewer" which you need to use to add the output to the code area.

Create a quiz game

A quiz game is fairly simple to implement. All you need to do is to add a few textboxes, assign the value you've entered to a variable and check whether your answer is correct.

Start with writing a title to your quiz using the show() method.

```
show("Welcome to the quiz")
```

Add title to the first question.

```
show("Question 1")
```

Write the some questions using ask() method and store them in variable named accordingly.

```
let question1 = ask("What is
1 + 1?")
let question2 = ask("What is
10 - 2?")
let question3 = ask("What is
5 * 2")
let question4 = ask("What is
100 / 1?")
```

This will store the answer to each question in the respective variables. You'll have to check whether the answers are correct. Do it using the 'if' statement.

```
if question1 == 2 {
     show("Question 1 is
correct")}
else {show("Question 1 is
wrong")}
```

Here we have checked whether the value stored in question1 is equal to 2. If it is, we will display the output as "Question 1 is correct". If it is any other value, we will display it using the else statement "Question 1 is wrong". You can do the same for other questions as well.

How about building a calculator using all the concepts you've already learnt? Challenge accepted!